Becoming Root

Although the Ubuntu security model has all administrative commands issued through sudo, there are some times when you really would be better off with a command prompt as root. Fortunately, there are many ways to accomplish this with sudo. A few examples:

```
sudo -s           # run a shell as root
sudo bash -o vi  # run a specific shell as root
sudo -i          # set root's initial login environment
sudo su - root    # become root and run root's login
environment
```

In the first two cases, the shell runs as root, but environment variables (for example, $HOME) are inherited. In the other two cases, the environment is replaced with root's real environment settings.

If you really need to be able to login as root, then you can use sudo passwd root to give the root account a password. As soon as you set the password, the root account can log in. You can always undo this change with sudo passwd -l root to lock the account.

Note

Enabling root logins is usually a bad idea. Logins give a trail of accountability. If someone logs in without a personalized account, then there is no way to identify the person who really logged in. sudo gives an audit trail in /var/log/auth.log (even though users with sudo access have the ability to blow away the logs).

Enabling Multiple CPUs (SMP)

Many of today's computers have multiple CPUs. Some are physically distinct, and others are virtual, such as hyper-threading and dual-core. In any case, these processors support symmetric multiprocessing (SMP) and can dramatically speed up Linux.

The kernel supports multiple CPUs and hyper-threading. If your computer has two CPUs that both support hyper-threading, then the system will appear to have a total of four CPUs.

There are a couple of ways to tell if your SMP processors are enabled in both the system hardware and kernel.

/proc/cpuinfo-This file contains a list of all CPUs on the system.

top-The top command shows what processes are running. If you run top and press 1, the header provides a list of all CPUs individually and their individual CPU loads. (This is really fun when running on a system with 32 CPUs. Make sure the terminal window is tall enough to prevent scrolling!)

System Monitor-The System Monitor applet can be added to the Gnome panels. When you click it, it shows the different CPU loads.